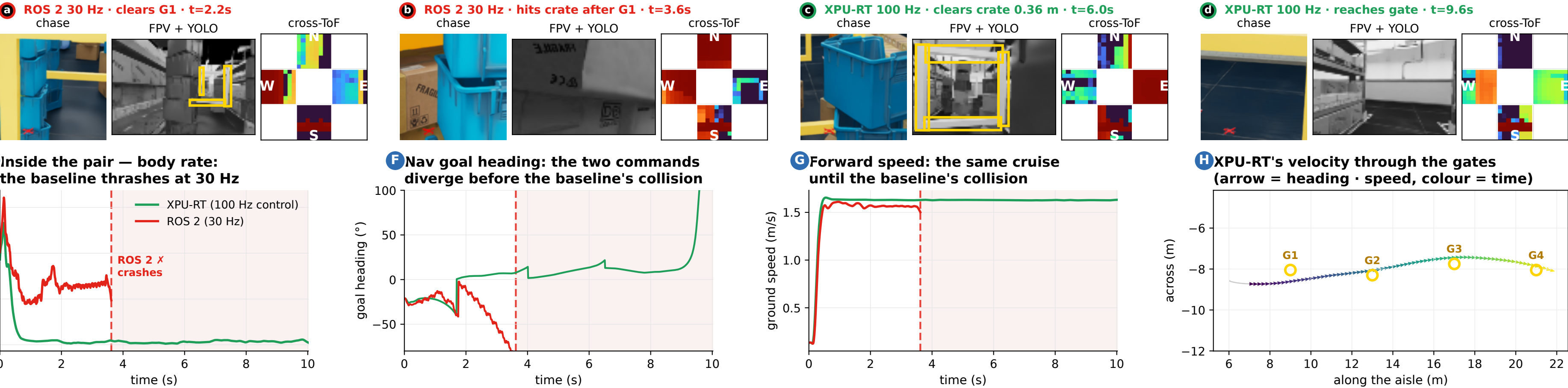
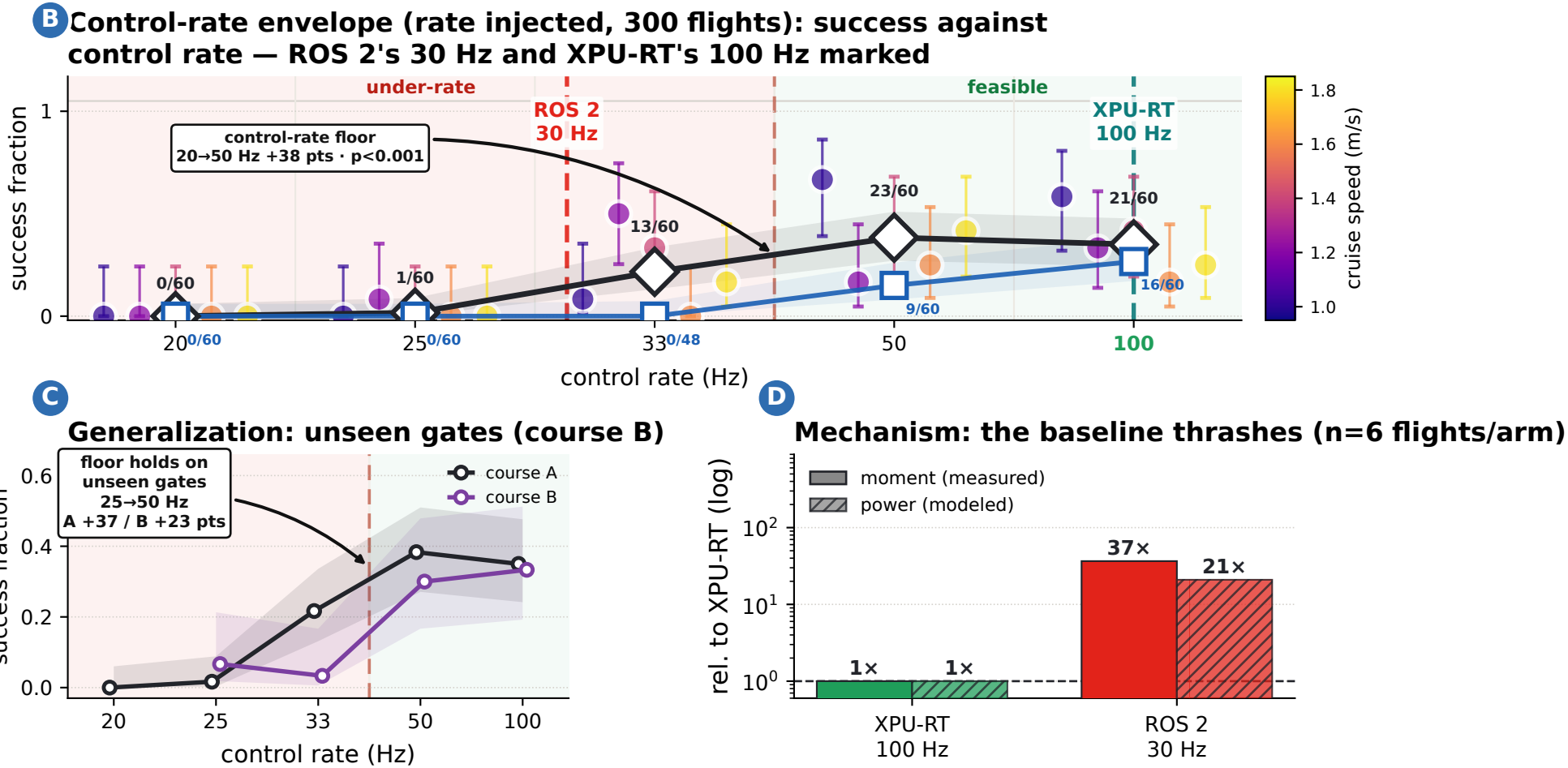
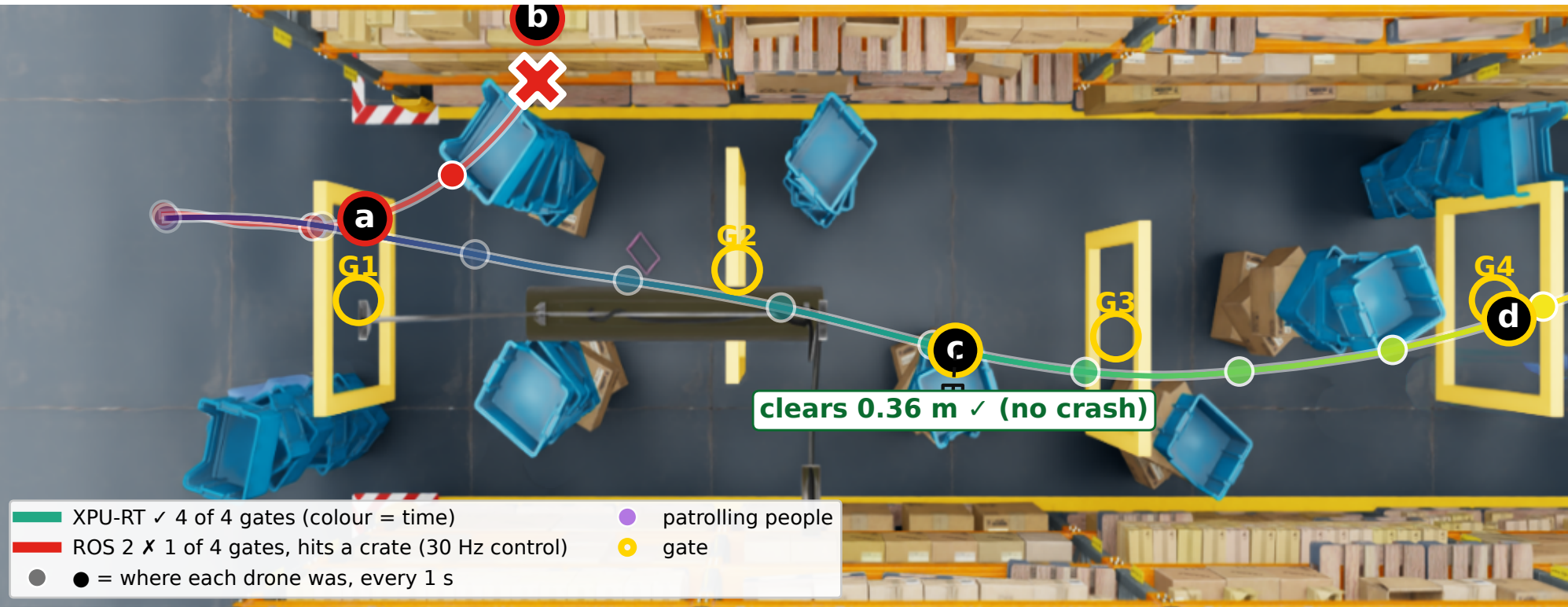


Warehouse gate-course showdown — same scene, drone and gain, at 1.6 m/s  
XPU-RT · solver-placed CP-SAT schedule (100 Hz control) clears all 4 gates  
ROS 2 · 4-hart YOLO pool, 4-wide nav, all 8 cores (30 Hz control) clears G1 → hits a crate  
one seed of this scene (seed 1001); over its 12 seeds neither arm completes — 2.3 vs 0.6 of 4 gates



**I** Onboard K1 schedule, measured at the 30 Hz camera — XPU-RT · solver-placed CP-SAT schedule places the kernels over all 8 harts: camera→control 27 ms, control every 10.0 ms, every frame on time · YOLO on 8 harts, 5 traced, nav on 8 harts, 3 traced, control on 1 hart  
ROS 2 · all 8 cores (3 processes · single executor · chained control · pinned): camera→control 30 ms, control every 33.4 ms, 0/598 frames late · YOLO on 4 harts, nav on 4 harts, 1 traced, control on 1 hart

